EVENT REPORT

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Training on FoodEx2

Parma, 12-13 April 2018

European Food Safety Authority

Abstract

A training was organised by the Evidence Management Unit of the European Food Safety Authority targeted to Member State representatives of the EFSA network on chemical occurrence data, project members of the EU Menu project and participants from the Food and Agriculture Organisation of the United Nations and the Tufts University Friedman School of Nutrition Science and Policy. The overall objectives of the training were to support the codification of foods according to FoodEx2, the food classification and description system of EFSA, provide guidance for the harmonised use of the system and the quality control of the codes. More specifically, this training was organised to enable participants to get familiar with the FoodEx2 catalogue browser, hierarchies and checking tool for the reporting of chemical occurrence, nutrient composition and food consumption data to EFSA. The training was evaluated positively by all participants. This event report summarises the background and rationale for the training organisation, its structure and main outcomes.

Key words: FoodEx2, codification, chemical occurrence, nutrient composition, food consumption

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1. Background, rationale and expected outcomes of the training

A training was organised by the Evidence Management Unit of the European Food Safety Authority (EFSA) targeted to Member State representatives of the network on chemical occurrence data, project members of the EU Menu project[1] and participants from the Food and Agriculture Organisation of the United Nations (FAO) and the Tufts University Friedman School of Nutrition Science and Policy (Tufts FSNSP), US. The overall objectives of the training were to support codification of foods according to FoodEx2, the food classification and description system of EFSA, and to provide guidance for the harmonised use of the system and the quality control of the codes, in view of its use for the reporting of data to EFSA and for the population of the Global Individual Food consumption data Tool[2] (GIFT) and Global Dietary database[3] (GDD) with food consumption data.

EFSA organised a dedicated training to enable participants to get familiar with the FoodEx2 catalogue browser, hierarchies and checking tool for the reporting of chemical occurrence, nutrient composition and food consumption data.

The specific objectives of this training were:

- To provide theoretical and practical support to participants in reporting data to EFSA according to FoodEx2;
- To harmonise and streamline reporting of chemical occurrence and food consumption data in order to ensure that the data collected are relevant and easy to analyse at the EU level;
- To promote best practise on the use of the FoodEx2 tools;
- To interact with competent organisations at global level for promoting the use of FoodEx2.

2. Participants and speakers

The audience consisted of 15 representatives from the EFSA chemical occurrence network (Belgium, Czech Republic, Denmark, Estonia, Ireland, Italy, Luxembourg, Portugal, Slovakia, Sweden, Norway and Turkey), five members of EU Menu project teams from three Member States (Italy, Poland and Romania), two colleagues from FAO, one representative from the University of Hohenheim, Germany, and one representative from the Tufts FSNSP.

The speakers at the training, Sofia Ioannidou and Tilemachos Goumperis, were representatives from EFSA.

See Appendix A for the list of participants and speakers.

3. Structure of the training

The training was structured as a two half-day sessions meeting and arranged in theoretical and practical sessions.

The agenda of the training is in Appendix B.

3.1. Opening of the training

The opening speech was given from Jeffrey Moon, leader of the EFSA Global Cooperation Team, and Mary Gilsenan, Head of Evidence Management Unit of EFSA. Jeffrey Moon welcomed all participants and highlighted the international interest FoodEx2 has attracted. He referred to the established collaboration EFSA has with FAO and the World Health Organisation (WHO) in the area of food consumption data sharing. He also referred to an upcoming collaboration agreement between EFSA and Tufts FSNSP, which facilitates the dissemination of FoodEx2 at a global level. Mary Gilsenan welcomed all participants and referred to the use of FoodEx2 at European level for the collection of data from different domains (i.e. chemical occurrence data, food consumption data through the EU Menu project, pesticides monitoring data etc.). She explained the training objectives and emphasised

[1] https://www.efsa.europa.eu/en/data/food-consumption-data
[2] http://www.fao.org/gift-individual-food-consumption/en/
[3] https://www.globaldietarydatabase.org/

EFSA's expectation to develop further FoodEx2 and update its catalogue in order to be able to capture foods that are consumed/analysed globally, and disseminate it further.

3.2. Introduction to FoodEx2

After a roundtable presentation, the trainers gave a general introduction to FoodEx2. They introduced the background for its development and its purpose. Food related information from different domains are routinely submitted to EFSA from different data providers, i.e. EU Member States' organizations, European Commission, Industry, Consumers associations, University, academia etc. In particular, FoodEx2 enables occurrence data to be combined with food consumption data in order to allow the calculation of dietary exposure, an important step of the risk assessment work carried out in EFSA. Special emphasis was given on the food groups and the facets present in the FoodEx2 catalogue and on how to use them. The FoodEx2 browser and the interpreting and checking tool were also introduced in detail.

For the purpose of the training the FoodEx2 browser, catalogue and interpreting and checking tool were made available through EFSA's Document Management System (DMS). For all external users the latest version of the FoodEx2 browser is available at GitHub, an open source software development platform[4]. The interpreting and checking tool is currently under revision and will shortly become available at GitHub.

3.3. Technical requirements

In order to facilitate the hands-on sessions, participants were asked to bring their own computer for the training where they needed to install the browser and the FoodEx2 catalogue either from the Document Management System (DMS) of EFSA or the open repository GitHub[4]. Support was provided on the installation and setup of the tools on the personal computers.

3.4. Theory and exercises

The theoretical part of the training provided basic knowledge on the use of FoodEx2 for reporting data based on the guidance of EFSA (EFSA, 2015). Attention was firstly focused on the starting point of thinking when codifying foods with FoodEx2. This is the 'What type of food is this?' which refers to the nature of the food itself and not its origin.

The three principle types of foods available in FoodEx2 are the raw primary commodities (RPC), the RPC derivatives and the composite food:

- RPCs are parts of plants physically separated after harvesting or animals after slaughtering, e.g. carrots. Processes that do not change the nature of food (e.g. freezing) can be applied
- RPC derivatives are obtained from raw commodities applying processes changing the 'nature' of food, e.g. flour after milling the grain
- Composite dishes are obtained by multiple commodities and/or derivatives/ingredients, after being processed and always involve recipes, e.g. ice-cream

Clear examples were provided for all three types of foods during the training.

3.5. Facets

An overview of the facets available in the system followed. The first focus was on the facets 'Source' (F01), 'Source commodities' (F27) and 'Ingredient' (F04). The decision on their use represents the answer to the question 'What type of food is this'. The facet 'Source' defines the origin of raw commodities and is usually already assigned as an implicit facet. The facet 'Source commodities' defines the origin of the derivative and the facet 'Ingredient' defines the origin of the composite food. Other facets were also described, such as the facet 'Process' (F28), 'Surrounding-medium' (F06), 'Sweetening-agent' (F08) etc. A demonstration on how to search for foods in the catalogue, select them, assign facets and copy either only the FoodEx2 code or the description as well, was given based on examples.

[4] https://github.com/openefsa/catalogue-browser/wiki

3.6. Special cases

The hands-on training proceeded with solutions on how to code special cases of foods as for example foods that are not already present in the list of base terms. Base terms are the foods appearing in the tree structure of the different hierarchies. These foods are not present either because they have not been considered so far for inclusion or they were deemed of minor relevance. In such cases the rule is to use the nearest upper level group specifying the additional information with facets; the facet F26.A07XE = generic term 'other' should also be added to communicate that the detailed term was not found in FoodEx2 and the description was built with facets. The following two scenarios for missing terms were discussed during the training:

- If a derivative is missing but the raw commodity is available. In this case it is advised to choose as a base term the nearest generic food group and assign under the source-commodities facet the suitable source raw commodity. As an example for this case the quinoa flour was given. Quinoa flour is not listed under the 'Cereal and cereal-like flours' (A04KS) but 'Quinoa grain' (A000R) does exist under 'Cereal grains (and cereal-like grains)'. Therefore, the FoodEx2 code for Quinoa flour can be built with facet descriptors and will look as follows: A04KS#F27.A000R$ F26.A07XE
- If a composite group is missing, it is advised to choose the correct 'branch' under the available composite dishes in FoodEx2; to choose a base term that is conceptually similar to the product being coded within this branch, provide the ingredients, add the facet F26.A07XE = generic-term 'other' and describe the food in a text field.

3.7. Mixed food products

Finally, the reporting of mixed food products was discussed. The coding of such foods depends on the nature of the components and their relative contribution. The following three cases were presented and examples and exercises were followed for each of them:

- In the case of mixed raw commodities or derivatives of the same nature, it is proposed to use a generic base term with multiple use of the 'Source-commodities' facet in order to allow the description of the presence of different raw commodities or derivatives. One of the examples discussed was the case of mixed dried fruits (banana and mangoes) for which the FoodEx2 code will be the A01QF#F27.A04JS$F27.A01LF (Mixed dried fruit, SOURCE-COMMODITIES = Bananas and similar-, SOURCE-COMMODITIES = Mangoes)
- In the case of mixed raw commodities or derivatives of substantially the same nature with minor ingredients, it is proposed to use a generic term, specify the components of the same nature with the 'source-commodities' facet and the 'foreign' component(s) with the 'ingredient' facet. One of the examples treated was the case of a bag of mixed salad with head lettuce, crisp lettuce, spinach leaves, beetroot leaves and carrots, for which the FoodEx2 code will be the A00KR#F04.A00QH$F27.A00KY$F27.A00MJ$F27.A00KZ$F27.A0DJS (Leafy vegetables, INGRED= Carrots, RACSOURCE= Head lettuces, RACSOURCE= Spinaches, RACSOURCE= Crisp lettuces, RACSOURCE= Beetroot leaves)
- In the case of mixed raw commodities or derivatives of different nature, it is proposed to consider the food as composite where a base term under the FoodEx2 composite dishes is selected and the components are listed as ingredients.

3.8. Regional food

The last part of the training was dedicated to codification of regional food items according to FoodEx2.

4. Evaluation of the training

Participants were invited to fill in a questionnaire survey to evaluate the training. Eight out of the 24 participants had previously followed training on FoodEx2. Feedback was received by all 24 participants. The results show that the participants have considered this training as very useful. It met their requirements and helped them to improve their knowledge on FoodEx2. The training was able to keep their interest at high levels and the time reserved for the exercises was considered sufficient.

The feedback includes also scores on different aspects of the training, which are summarised in Appendix C.

5. Conclusions

During the training on FoodEx2, organised by the Evidence Management Unit of EFSA, guidance was provided to the participants for the harmonised use of the system and the quality control of the codes through theory and exercises. The participants advanced their knowledge on the use of FoodEx2, learned to select the correct base term for the food they need to report and specify its characteristics with facet descriptors. They also got acquainted with the FoodEx2 catalogue browser and interpreting and checking tool. The training was evaluated positively by all participants. They appreciated the fact that specific questions and difficulties could be addressed directly by the trainers in person. The theory of FoodEx2 was well perceived in addition to the interactive work, clarity and comprehensiveness of the exercises. The training was also a means to interact with competent organisations at a global level for promoting the use of FoodEx2. The organisation of webinars on FoodEx2 is envisaged for the near future in order to facilitate the participation of more people.

References

EFSA, 2015. The food classification and description system FoodEx 2 (revision 2). EFSA Supporting Publications. 2015;12(5):EN - 804. 90 pp.

Abbreviations

DMS	Document Management System
EFSA	European Food Safety Authority
EU	European Union
FAO	Food and Agriculture Organisation of the United Nations
GDD	Global Dietary database
GIFT	Global Individual Food consumption data Tool
RPC	Raw Primary Commodities
US	United States
Tufts FSNSP	Tufts University Friedman School of Nutrition Science and Policy
WHO	World Health Organisation

Appendix A – List of participants and trainers

Participants		Country	Institution
Vromman	Valerie	Belgium	Federal Agency for the Safety of the Food Chain Control Policy
Rehurkova	Irena	Czech Republic	National Institute of Public Health
Procházková	Jana	Czech Republic	National Institute of Public Health
Bjorn Petersen	Pernille	Denmark	National Food Institute Technical University of Denmark
Padur	Kadi	Estonia	Veterinary and Food Board
Eleraky	Laila	Germany	University of Hohenheim
Karageorgou	Dmitra	Greece	Tufts University
Stack	Martina	Ireland	Food Safety Authority of Ireland
De Martino	Michele	Italy	Istituto Superiore di Sanità - Ministero della Salute
Durazzo	Alessandra	Italy	Istituto Nazionale di Ricerca per gli Alimenti e la Nutrizione
Balcerzak	Agnieszka	Italy	Food and Agriculture Organisation of the United Nations
Padula De Quadros	Victoria	Italy	Food and Agriculture Organisation of the United Nations
Zust	Danny	Luxemburg	Ministère-Direction de la Santé
Al-Qaisy Ahmed	Waleed Saleh	Norway	Norwegian Food Safety Authority
Oltarzewski	Maciej	Poland	National Food and Nutrition Institute
Matczuk	Ewa	Poland	National Food and Nutrition Institute
Oliveira	Luisa	Portugal	National Institute of Health
Tanasescu	Bogdan	Romania	National Sanitary Veterinary and Food Safety Authority
Bojan	Ioan	Romania	National Sanitary Veterinary and Food Safety Authority
Svetlikova	Angela	Slovakia	Food Safety Authority- Ministry of Agriculture and Rural Development
Fogelberg	Petra	Sweden	National Food Administration
Foster	David	Sweden	National Food Administration
Serdaroglu	Fatih	Turkey	The Ministry of Food, Agriculture and Livestock
Cicek	Kenan	Turkey	The Ministry of Food, Agriculture and Livestock
Trainers			**Institution**
Ioannidou	Sofia		European Food Safety Authority
Goumperis	Tilemachos		European Food Safety Authority

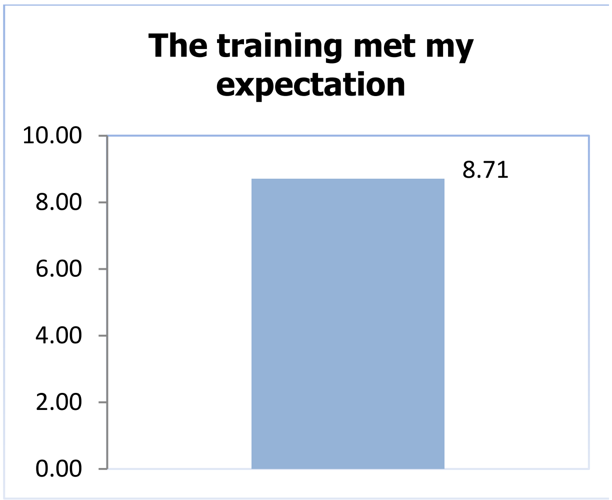
Appendix B – Agenda

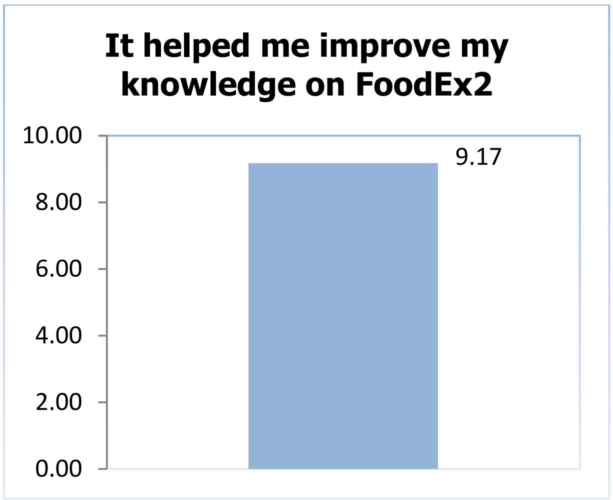
12th April 2018, Chair: Sofia Ioannidou, Scientific Officer, Evidence Management (DATA) Unit, EFSA	
Time	***Items***
14.00 – 14.15	**Welcome address by EFSA**
14.15 – 14.30	Roundtable presentation
14.30 – 16.00	**General introduction to FoodEx2** Why it was developed Its purpose Content of FoodEx2 and how to use FoodEx2 browser Check installation of FoodEx2 browser in participants' computers
16.00 – 16.30	Coffee break
16.30 – 17.45	**Theory and Exercises** Raw commodities Derivatives Composite dishes
17.45 – 18.00	Introduction to the documentation and tools supporting the use of FoodEx2
18.00	**Closure of the 1st day**

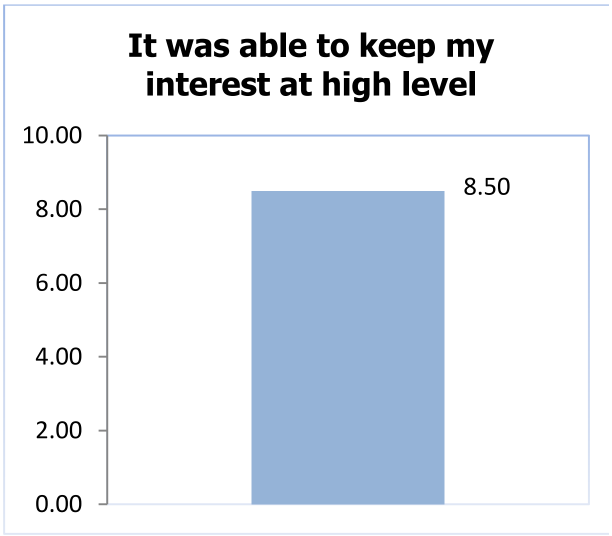
13th Aril 2018, Chair: Sofia Ioannidou, Scientific Officer, Evidence Management (DATA) Unit, EFSA	
Time	***Items***
8.30 – 8.35	**Welcome**
8.35 – 10.30	**Theory and exercises** Facets and facet descriptors
10.30 – 11.00	Coffee break
11.00 – 13.00	**Special cases** How to codify regional food items in FoodEx2 format
13.00	**End of the training**

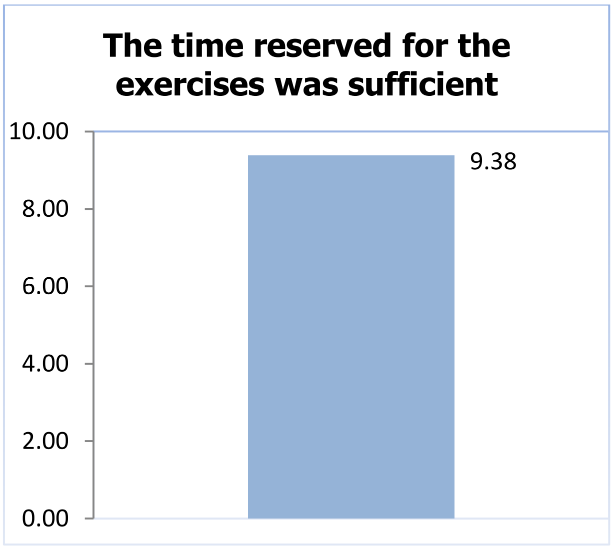
Appendix C – Training evaluation

Quantitative evaluation (the results as average scores are displayed graphically below):









Qualitative evaluation (selected answers)

1. **Which aspects you liked the most in the FoodEx2 training performed during these two days?**

- Interactive/practical work
- Clarity and comprehensiveness of the exercises
- Simple, clear overview of the theory of FoodEx2, getting familiar with the concept of facet descriptors and how to use them
- Using FoodEx2 for coding composite dishes

2. **What were the benefits you received from the training?**

- Better understanding of FoodEx2 system and its rules, a lot of practical work that made people feel confident in using it
- The in person, face-to-face training was very beneficial, as specific questions and difficulties could be addressed

3. **What should be included in future trainings on FoodEx2?**

- More practical work
- More examples on other data flows as food contact material and veterinary drugs
- Specific examples on codification of dietary supplements
- A small section that will include specific questions from data managers on the codification of local foods
- Mapping with SSD2; more information on how to integrate the tool in data submissions

4. **Suggestions/proposals**

- Thanks to EFSA for the organisation and implementation of the training
- Organise webinars that will be reachable by more people
- The term naming and definition tab should be made available on the facet screen of the browser